SAR Target Classification Using the Multikernel-Size Feature Fusion-Based Convolutional Neural Network

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***Abstract*— It is well-known that the convolutional neural network (CNN) is an effective method for synthetic aperture radar (SAR) target classification. In the convolutional layer of CNN, convolutional kernels of different sizes can extract different feature information of the target. The small-size kernel can extract the local texture feature information, and the large-size kernel can extract the global contour feature information. Traditional CNN methods usually use fixed-size kernels for convolution, and they generally lose part of the target's feature information, resulting in the inaccurate classification of the SAR targets. This article proposes a novel CNN model based on multikernel-size feature fusion (MKSFF-CNN) for SAR target classification. MKSFF-CNN designs a convolutional methodology with a multichannel parallel topology, it uses convolutional kernels of different sizes to extract the multikernel-size deep features of the SAR target, and then, these features are fused in an optimal way to acquire the lowest loss. Moreover, MKSFF-CNN concatenates the fused features extracted by the convolutional layers of different dimensions to achieve the finest classification. MKSFF-CNN greatly elevates the feature representation completeness of the SAR targets so that more useful feature information can be exploited for SAR target classification. Undoubtedly, MKSFF-CNN can achieve a better classification performance compared with traditional CNN models with a fixed kernel size. The superiority of MKSFF-CNN is validated on the moving and stationary target acquisition and recognition (MSTAR) dataset with the detailed objective and subjective evaluation.**

***Index Terms*— Convolutional neural network (CNN), multidimensional feature concatenation, multikernel-size-based convolution, optimal feature fusion, synthetic aperture radar (SAR) target classification.**

I. Introduction

SYNTHETIC aperture radar (SAR) is an active imaging sensor with all-day, all-weather capability, and it acquires high-resolution remote sensing images [1], [2]. Compared to optical images, SAR images can provide much more useful information, and they can be effectively used in the fields of military reconnaissance, marine surveillance, and so on. Among them, SAR target classification is an important application in the field of military reconnaissance, and it has received great attention and becomes a research hot topic [3].

A standard architecture of the SAR automatic target recognition (SAR-ATR) system proposed by the MIT Lincoln Laboratory is composed of three stages: detection, discrimination, and classification [4]. Detection is used to prescreen the regions of interest (ROIs) according to the local contrast information between the targets and the surrounding clutter [5]. The ROIs might include not only targets but also false alarms. At the discrimination stage, certain discriminators are designed to eliminate the false alarms and reserve the real targets [6]. Finally, the discriminated targets are categorized into different types.

Among all the stages of SAR-ATR, the classification is so important for military reconnaissance that it attracts great intention. According to the different classification theories, the SAR target classification methods can be generally divided into three categories: template-based methods [7], model-based methods [8], and machine-learning-based methods [9]. The core of the template-based methods is the design of the matching templates. For example, in the semiautomated image intelligence processing (SAIP) system [7], different target characteristics are obtained, and then, these target characteristics are input into the mean-square-error classifier to find the best match between the SAR target data and the template database. The classification performance of the SAIP system is acceptable if the SAR target is similar to the designed template; otherwise, the classification accuracy degrades significantly [10]. In addition, the establishment of a template database consumes a lot of time so that the template-based methods are not well promoted in practical applications.

Compared with the template-based methods, the model-based methods use a more sophisticated process. The moving and stationary target acquisition and recognition (MSTAR) system [11] first produces a set of hypotheses concerning category, attitude, and so on of the SAR image chips; then, it uses 3-D modeling and electromagnetic calculation to simulate the image chip of the SAR target that meets the relevant hypotheses. Finally, the features of these simulated SAR images are continuously compared with those of real

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SAR images to adjust the model until the output has achieved a sufficiently high classification confidence level.

In recent years, with the continuous progress of machine learning theory, more and more SAR target classification methods based on machine learning have been proposed. The SAR target classification methods based on machine learning usually include two stages: feature extraction and classification [12]. There are many features specifically designed for the SAR target classification, such as the shape feature [13], texture feature [14], edge feature [15], electromagnetic scattering feature [16], saliency feature [17], saliency plus the gist feature (SG feature) [18], and semantic feature [19]. Moreover, based on these features, various feature extraction operators are designed, such as histograms of oriented gradients (HOG) [20], scale-invariant feature transform (SIFT) [21], speeded up robust features (SURF) [22], local binary pattern (LBP) [23], completed local binary pattern (CLBP) [24], and haar-like [25]. After the feature extraction procedure, the extracted features are input into a certain classifier for the final decision. Common classifiers mainly include k-nearest neighbor (KNN) classifier [26], support vector machine (SVM) classifier [27], Adaboost classifier [28], and so on. The target classification performance mainly relies on the feature representation capability. When the extracted feature is unable to highlight the difference between different types of targets, the classification performance deteriorates sharply.

Deep learning [29] goes a step further on the basis of traditional machine learning methods, and it combines the feature extraction stage and the classification stage in one model. LeCun *et al.* [30] proposed a convolutional neural network (CNN). CNN can automatically extract the deep features of the images, and it achieves a better performance on SAR target classification [31]. CNN usually uses the softmax classifier for classification, but the performance of the softmax classifier decreases when the number of positive and negative samples differs greatly. CNN-SVM [32] and CNN-Adaboost [33] replace the softmax classifier with the SVM classifier and Adaboost classifier, respectively. However, these models are not competent for complicated classification tasks due to their shallow depth. A-ConvNet [12] uses the convolutional layer instead of the fully connected layer for data dimensionality reduction and increases the depth of the network. Visual geometry group network (VGG-Net) [34] uses stacked small-size kernels instead of single large-size kernels, making the depth of the model increase. The residual neural network (Res-Net) [35] designs a residual unit to further increase the depth of the network. The generative adversarial network (GAN) [36] adopts a new structure to increase the complexity of the model. These improved models get higher classification accuracy, but they all ignore the local texture and edge information (low-level features), which are of great significance for SAR target classification. Part-based convolutional neural network (PBNet) [37] generates more features by aggregating local context information and global context information and enhances the ability of feature representation. Convolutional highway unit network (CHU-Net) [38] increases the width of the network by combining a convolutional highway unit with 3×3 kernels and a convolutional highway unit with 5×5 kernels, thereby extracting more features of the SAR target. Deformable contextual and boundary-weighted network (DABNet) [39] proposes a deformable context feature pyramid module and a boundary-weighted loss function to improve the completeness of multiscale feature representation. Multiscale rotation-invariant haar-like feature integrated CNN (MSRIHL-CNN) [40] and deep CNN with multiscale rotation invariant completing local binary patterns feature (CNN + CLBP) [41] use haar-like templates and CLBP templates to extract the low-level edge and texture features ignored by CNN networks, respectively. Then, the low-level edge and texture features and the deep features extracted by CNN networks are optimally fused to acquire a better classification. As the complexity of the model becomes higher and higher, the sample size and time required in the training process are also significantly increased. In order to deal with the classification dilemma of limited sample size, a lot of CNN methods based on sample expansion [42]–[45] have been proposed. Among them, the multiview deep learning framework (M-VDCNN) [45] not only proposes a sample expansion scheme based on multiple views but also designs a parallel network topology with multiple inputs. In addition, few-shot learning [46], metalearning [47], and transfer learning [48] simplify the training process through model transfer between tasks so that they can perform well on small samples. However, almost all of these CNN methods only use the kernels of a fixed size for convolution, resulting in some feature representation loss of the SAR targets.

Since traditional networks cannot achieve a promising feature representation completeness as a result of the fixed-kernel-size-based convolution, this article proposes a novel CNN model based on multikernel-size feature fusion (MKSFF-CNN) for SAR target classification. MKSFF-CNN uses convolutional kernels of different sizes to extract the multikernel-size features of the SAR target, and then, these features are fused in an optimal way to minimize the loss. Moreover, MKSFF-CNN concatenates the fused features extracted by the convolutional layers of different dimensions to achieve the finest classification. The main contributions of this article are threefold.

1) MKSFF-CNN proposes a new network with a convolutional methodology of multisized kernels. MKSFF-CNN uses kernels of different sizes for convolution to obtain the multikernel-size deep features, and these features are fused in an optimal way to acquire a minimum loss. Moreover, the kernels of different sizes are obtained through dilation so that the number of parameters that need to be updated can be greatly reduced. MKSFF-CNN significantly improves the SAR target feature representation completeness through the multikernel-size-based convolution, and it can achieve high efficiency.
2) MKSFF-CNN concatenates the edge and texture features extracted by the shallow convolutional layer and the complex abstract features extracted by the deep convolutional layer, and they all participate in the classification decision stage with optimal weighting coefficients automatically generated by the network training so that the SAR target feature representation completeness is further improved.

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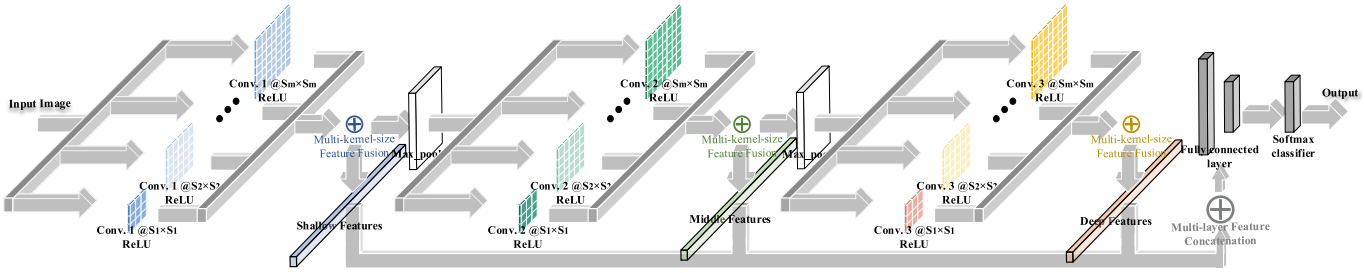


Fig. 1. Basic architecture of the proposed MKSFF-CNN. S_1, S_2, …, S_m: the different sizes of the convolutional kernel.

3) Both the fusion of the multikernel-size features and the concatenation of the multilayered features are realized through automatic network training. The optimal weights used in fusion and concatenation are automatically obtained in the case of minimizing the classification error. The training process without human intervention allows the model to be updated with the best state, so the classification accuracy can be greatly improved.

The structure of the article is organized as follows. Section II gives a detailed introduction of the SAR classification methodology. Section III is the experimental part, and the classification performance is validated on the MSTAR dataset with a detailed evaluation. Finally, Section IV gives the conclusion of the article.

II. MKSFF-CNN Classification Methodology

The main structure of the proposed MKSFF-CNN is shown in Fig. 1, where S_1, $S_2, \ldots, S_m$ are the different sizes of the convolutional kernels used in each convolutional layer, and m is the number of the convolutional kernels with different sizes. MKSFF-CNN uses a convolutional layer with a multichannel parallel topology. This special convolutional layer uses kernels of different sizes to perform convolution simultaneously, so MKSFF-CNN can completely extract the multikernel-size feature information of the SAR targets. In addition, MKSFF-CNN concatenates the features extracted by the convolutional layers of multiple dimensions and inputs them into the fully connected layer so that more essential feature information can be used for the final classification.

The backbone of the proposed MKSFF-CNN includes three convolutional layers, two pooling layers, two fully connected layers, and a softmax classifier. When the SAR image chip is input, the multikernel-size feature information of the input data is parallel extracted in the convolution layers, and the extracted multikernel-size features are fused in an optimal way to minimize the loss. Finally, the fused multikernel-size features of different dimensions are concatenated and fed into the first fully connected layers. The fully connected layers and the softmax classifier constitute the decision-making part, which assigns appropriate posterior probabilities to each target type to achieve the final classification result. The detailed methodology of the proposed MKSFF-CNN is described in the following parts of this section.

A. Convolution and Nonlinearization

The convolutional layer is the core of the proposed MKSFF-CNN structure. It can automatically extract the deep features and plays an essential role in the fine classification of SAR targets. In the convolutional layer, the feature information extracted by convolutional kernels of different sizes is not the same. The small-size kernel can extract the local texture feature information, and the large-size kernel can extract the global contour feature information. Traditional CNN methods usually use fixed-size convolutional kernels for convolution, so they generally lose some feature information of the target, resulting in unsatisfactory classification results. Fig. 2 shows the SAR target feature representation performance obtained by convolution with kernels of different sizes in the form of the feature map. It can be seen in Fig. 2 that the target in the feature map is scattered when the 3×3 kernel is used so that the global contour feature of the target is missing, and a large amount of speckle noise in the background is retained. When the 7×7 kernel is used, the details such as the local texture of the target are lost. When the 5×5 kernel is used, the global contour feature and the local texture feature of the target are well preserved, but the speckle noise that is close in the background is connected to produce striped noise after convolution, and it will be misjudged as target feature information, causing misclassification. To solve this problem, the proposed MKSFF-CNN proposes a convolutional layer with a multichannel parallel topology. MKSFF-CNN utilizes different sizes of kernels to perform convolutional operations on the input SAR image simultaneously to extract multikernel-size features, and these multikernel-size features are fused with optimal weights acquired by network training. It can be found from Fig. 2 that, after the optimal fusion of multikernel-size features, both the global contour feature information and the local texture feature information of the target in the feature map are greatly preserved, and the speckle noise in the background is better suppressed at the same time.

In the nth convolutional layer of the proposed MKSFF-CNN, $f_i^{(n-1)}$ is supposed as the ith input feature map of this layer, and it is connected to all of the outputs of this layer. $f_{j_m}^{(n)}$ is assumed as the jth output feature map of the mth convolution channel of this layer. Let $W_{ij_m}^{(n)}$ be the convolution kernel that operates $f_i^{(n-1)}$ to $f_{j_m}^{(n)}$ in the nth convolutional layer, and $b_{j_m}^{(n)}$ is the deviation of $f_{j_m}^{(n)}$; then, the specific forward propagation process in the convolution process can

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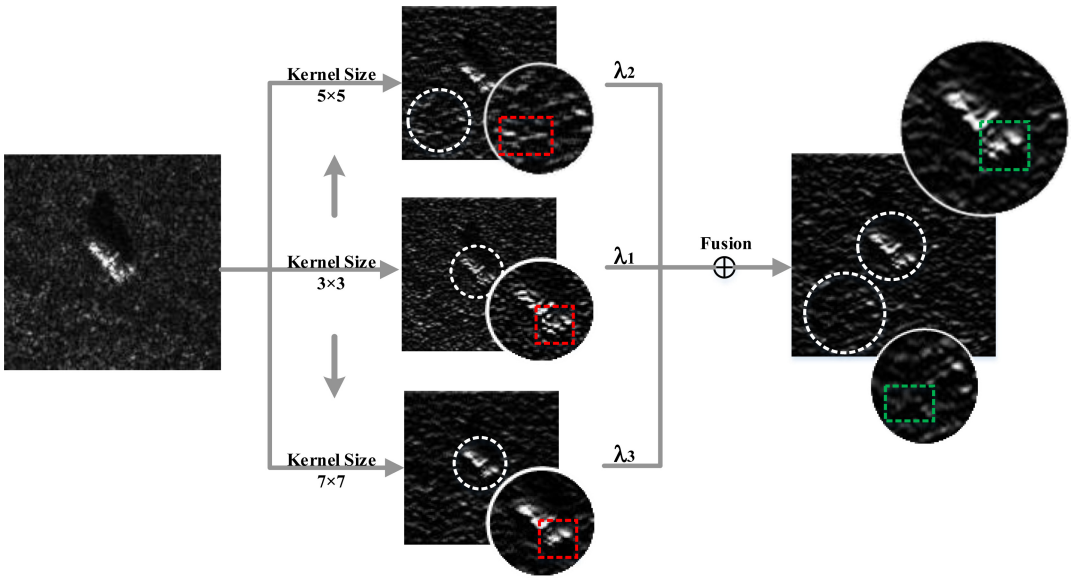


Fig. 2. Better SAR target feature representation performance of the proposed MKSFF-CNN.

be expressed as

$$\alpha_{j_m}^{(n)} = \sum_i \left(f_i^{(n-1)} * W_{ij_m}^{(n)} + b_{j_m}^{(n)} \right) \tag{1}$$

$$f_{j_m}^{(n)} = \sigma\left(\alpha_{j_m}^{(n)} \right) \tag{2}$$

where $\alpha_{j_m}^{(n)}$ represents the convolution result before nonlinear activation. The symbol $*$ represents the convolution operation, and $\sigma(\cdot)$ is the nonlinear activation function. The purpose of the nonlinear activation operation is to improve the nonlinear characteristics of the network. The proposed MKSFF-CNN uses the rectified linear unit (ReLU) [49] function as the nonlinear function. The ReLU function can avoid problems, such as gradient explosion and gradient disappearance, when the input is larger than 0. In addition, because ReLU does not need to perform exponential calculations, the calculation cost of the network is significantly reduced. The ReLU can be formulated as

$$\sigma\left(\alpha_{j_m}^{(n)} \right) = \max\left\{ 0, \alpha_{j_m}^{(n)} \right\}. \tag{3}$$

After extracting the multikernel-size features, the network will fuse these features optimally. The specific process is

$$f_j^{(n)} = \sum_m \lambda_m^{(n)} \cdot f_{j_m}^{(n)} \tag{4}$$

$$\lambda_1^{(n)}, \lambda_2^{(n)}, \ldots, \lambda_m^{(n)} = \arg\min\left(E^{(n)} \right) \tag{5}$$

where $E^{(n)}$ is the loss of the nth convolutional layer. $\lambda_m^{(n)}$ is the fusion weight of the mth convolutional channel in the nth convolutional layer, and the fusion weights are constantly updated during the network training process to minimize the loss. The function of $\arg\min(\cdot)$ represents the variable values when the objective function is minimized, and $f_j^{(n)}$ represents the jth fused feature in the nth convolutional layer.

Since the 1×1 kernel has no help in increasing the receptive field, it is not considered. The 9×9 kernel is so large that it repeatedly extracts the feature information when it slides on the SAR image with a small stride, resulting in the decrease of the completeness of the feature representation. Therefore, in the proposed MKSFF-CNN, the number of convolutional kernels with different sizes in each convolutional layer is set to three, and S_1, S_2, and S_3 are set to three, five, and seven, respectively, so as to improve the feature representation completeness to the most extent. Since the increase in the kernel size will cause the computing time to increase exponentially, the proposed MKSFF-CNN obtains the other two sizes of kernels by dilating the 3×3 kernel so that the computational burden is greatly reduced. For example, a 5×5 kernel obtained through dilation has the same local receptive field compared to the ordinary kernel of the same size, but it only consumes less than half of the calculation time and memory. In addition, because of the relationship among kernels of different sizes, the features extracted through different convolution channels are mainly different in the receptive field, while the information they represent has a certain connection. Therefore, they are complementary in the feature fusion process, so as to better characterize the local texture information and global contour information of the target, thereby ensuring the improvement of the completeness of the feature representation.

B. Pooling

The pooling layer is generally used to reduce the dimensionality of the feature map, thereby reducing the parameters of the fully connected layer and further reducing the computational burden of the entire network. The proposed MKSFF-CNN uses maximum pooling as the pooling operator, which selects the maximum value of a set of data in the local window as the output. Maximum pooling can achieve a certain degree of invariance, including translation invariance, rotation invariance, and scale invariance. The specific maximum pooling process is

$$f_j^{(n)}(x, y) = \max_{0 \le p < w_1, 0 \le q < w_2} f_j^{(n)}(x + p, y + q) \tag{6}$$

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where (x, y) is the coordinates. w_1 and w_2 are the sizes of the local window of the pooling layer. A lot of research works have shown that a larger pooling size is likely to cause more information to be lost in the downsampling process, thereby affecting the subsequent classification accuracy, so the pooling size is generally set to be relatively small. The pooling size of the proposed MKSFF-CNN is set to 2×2.

C. Fully Connected Layer and Dropout

The fully connected layer generally plays the role of information extraction and integration in the network. Its essence is equivalent to the spatial transformation of features. It reduces the parameters of the network and enhances the nonlinearity combined with the nonlinear mapping of the activation function.

After the input SAR image is subjected to convolution and pooling operations, the feature information is extracted. The feature information will be converted into feature vectors and input into the fully connected layer. Traditional CNN generally only inputs the abstract features extracted by the last-dimensional convolutional layer into the fully connected layer, while the proposed MKSFF-CNN concatenates the concrete and abstract fused features extracted from the convolutional layers of different dimensions, and then, it inputs the concatenated fused feature vector to the fully connected layer. Let $F^{(n)} = [f_1^{(n)}, f_2^{(n)}, \ldots, f_{K_n}^{(n)}]$ be the fused feature extracted by the nth convolutional layer, and K_n is the number of features extracted by the nth convolutional layer. Then, the fused features extracted by the convolutional layers of multiple dimensions are concatenated to obtain the concatenated fused feature vector of F

$$F = \left[\omega_1 \cdot F^{(1)}, \omega_2 \cdot F^{(2)}, \ldots, \omega_C \cdot F^{(C)}\right] \quad (7)$$

$$\omega_1, \omega_2, \ldots, \omega_C = \arg\min(E) \quad (8)$$

where E is the cost function. $\omega_1, \omega_2, \ldots, \omega_C$ are the optimal concatenated weights, which are constantly updated during network training to maximize the subsequent classification accuracy. C is the number of fused features used for concatenation, and it is also the number of the convolutional layers in the network. After the concatenated fused feature vector is obtained, the flattening operation is performed on it to obtain the flat concatenated fused feature vector of F, which will be input into the fully connected layer

$$V = \sigma(\mathcal{F} \cdot \mathcal{W} + \mathcal{B}) \quad (9)$$

where $\mathcal{W}$ and $\mathcal{B}$ are the weight and deviation of the fully connected layer, respectively. V is the output of the fully connected layer. Assuming that $\mathcal{F} \in \mathbb{R}^C$ and the dimension of V is L, the size of $\mathcal{W}$ is $C \times L$, $\mathcal{B} \in \mathbb{R}^L$, and $V \in \mathbb{R}^L$.

When there are fewer samples in the training set, it will lead to overfitting problems in the complex neural network during the training process, making the model perform very badly on the test set, or even unusable. In order to effectively solve this problem, dropout is used in deep learning. Dropout sets a parameter of dropout ratio, which is used as the probability of setting each hidden unit to 0. In each training process, different hidden units are randomly set to 0, and only the relevant parameters of the reserved hidden units are updated. This random composition process of the network structure strengthens the generalization ability of the model. The dropout can be used in both the convolutional layer and the fully connected layer. In the proposed MKSFF-CNN, since the parameters of the fully connected layer are far more than the convolutional layer, dropout is used in the fully connected layer, and the dropout ratio is set to 0.5.

D. Softmax

The softmax classifier is often used for multitarget classification. It generally maps the output of the fully connected layer to the (0, 1) interval to provide posterior probabilities for each category. Let the output of the fully connected layer be $V = [V_1, V_2, \ldots, V_L]$; the corresponding posterior probabilities for each class can be formulated as

$$p(y = y_i|V) = \frac{\exp(V_i)}{\sum_{j=1}^{L} \exp(V_j)} \quad (10)$$

where y and y_i represent the category of the current target and the ith category in the multiobject classification task. The output of the softmax classifier is also an L-dimensional vector, whose dimension is the number of categories in the multitarget classification task, and the elements are the probability that the current target belongs to each category. Generally, in the output vector of the softmax classifier, the category corresponding to the largest probability is the category to which the current target belongs.

E. Cost Function and Backpropagation

The cross entropy is applied as the cost function of the proposed MKSFF-CNN; it is formulated as

$$E(\boldsymbol{W}, \boldsymbol{b}) = -\sum_{i=1}^{L} y_i \cdot \log p(y = y_i|V; \boldsymbol{W}, \boldsymbol{b}) \quad (11)$$

where $\boldsymbol{W}$ and $\boldsymbol{b}$ are the parameters that can be updated in the network. These trainable parameters are optimized by continuously minimizing the cost function during the training process so that the final classification accuracy can be continuously improved.

In the specific network training phase, backpropagation is used to calculate the gradient and update the parameters

$$\boldsymbol{W} \leftarrow \boldsymbol{W} - \mathcal{R} \cdot \frac{\partial E}{\partial \boldsymbol{W}}, \quad \boldsymbol{b} \leftarrow \boldsymbol{b} - \mathcal{R} \cdot \frac{\partial E}{\partial \boldsymbol{b}} \quad (12)$$

where $\mathcal{R}$ is the learning rate.

In the process of backpropagation [50], the error term of each layer and unit is generally calculated first, and then, the partial derivatives of the cost function can be further calculated. For different layers, the error term calculation method is different. The error term for the output layer is computed as

$$\delta_i^{(V)} = p(y = y_i|V) - y_i \quad (13)$$

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where $\delta_i^{(V)}$ represents the error term of the ith unit in the output layer, and it can be fed back to the lower layer to calculate the corresponding error term.

In the convolutional layer of the proposed MKSFF-CNN, the error term of the nth layer is calculated by the convolution kernel and the error term of the $(n+1)$th layer

$$\delta_i^{(n)} = \sum_j \left[\delta_j^{(n+1)} * \mathrm{ROT}_{180^\circ}\left(W_{ij}^{(n+1)}\right)\right] \odot \sigma'\left(\alpha_i^{(n)}\right) \quad (14)$$

where $\mathrm{ROT}_{180^\circ}(\cdot)$ refers to the operation of 180° rotation of the matrix. $\odot$ represents the Hadamard product, i.e., the operation of multiplying the corresponding position elements, and $\sigma'(\cdot)$ is the derivative of the nonlinear activation function.

In the pooling layer of the proposed MKSFF-CNN, the error term of the nth layer is obtained by

$$\delta_i^{(n)} = \mathrm{upsample}\left(\delta_i^{(n+1)}\right) \odot \sigma'\left(\alpha_i^{(n)}\right) \quad (15)$$

where upsample$(\cdot)$ denotes the up-sampling operation, which returns all the values in the current layer to the corresponding positions of the previous layer and, at the same time, zeroes other positions in the previous layer.

After the error term of each layer and unit is got, the partial derivatives of the cost function can be given

$$\frac{\partial E}{\partial W_{ij}^{(n)}} = \mathrm{ROT}_{180^\circ}\left(f_i^{(n-1)} * \mathrm{ROT}_{180^\circ}\left(\delta_j^{(n)}\right)\right) \quad (16)$$

$$\frac{\partial E}{\partial b_j^{(n)}} = \sum_{x,y} \delta_j^{(n)}(x, y). \quad (17)$$

Then, the network updates the parameters through (16) and (17) and repeats the above operation after a new SAR image is input.

So far, the overall framework and calculation process of the proposed MKSFF-CNN have been described.

III. Experiments and Analysis

A. Datasets for Experiments

The SAR image dataset used in this article is from the MSTAR program supported by the U.S. Defense Advanced Research Projects Agency (DARPA) and the U.S. Air Force Research Laboratory. The dataset is obtained near Huntsville, Alabama, by the STARLOS sensor, of which the resolution is 0.3 m, the band is X, and the polarization is horizontal–horizontal (HH). The MSTAR dataset contains ten types of ground targets captured under different conditions (such as aspect angle, depression angle, and serial number). They are T-62 and T-72 belonging to tank, 2*S*1 belonging to the rocket launcher, ZIL-131 belonging to the truck, BTR-60, BTR-70, BRDM-2, and BMP-2 belonging to armored personnel carrier, ZSU-234 belonging to air defense unit, and D7 belonging to the bulldozer. There is generally only one target located in the image center of the MSTAR dataset. The MSTAR dataset is widely used in the performance comparison of SAR-ATR algorithms, and the optical images of ten types of ground targets and the corresponding SAR images are shown in Fig. 3.

TABLE I
Number of Training and Test Images for the SOC Experimental Setup

Types	Training set		Test set	
	Depression	Number	Depression	Number
2S1	17°	299	15°	274
BRDM-2	17°	298	15°	274
BTR-60	17°	256	15°	195
D7	17°	299	15°	274
T-72	17°	232	15°	196
BMP-2	17°	233	15°	195
BTR-70	17°	233	15°	196
T-62	17°	299	15°	273
ZIL-131	17°	299	15°	274
ZSU-234	17°	299	15°	274

In order to comprehensively evaluate the classification performance of the proposed MKSFF-CNN, the standard operating condition (SOC) and the depression-variant extended operating condition (EOC-D) were used to test the algorithm.

The SOC uses ten types of targets in the MSTAR dataset for classification. SAR image data with a depression angle of 17° are selected as the training set, and SAR image data with a depression angle of 15° are selected as the test set, as shown in Table I. In addition, no matter in the training set or the test set of the SOC, the sample size of each type of target is not exactly the same.

The SAR images of the same target generally have large differences under different acquisition conditions. Especially when the depression angle is different, this situation is particularly serious. Therefore, the classification test of the proposed MKSFF-CNN in EOC-D is of great significance to the objective evaluation of the algorithm. As shown in Table II, EOC-D selects 2*S*1, BRDM-2, T-72, and ZSU-234 for classification. The SAR images for training are captured at a 17° depression angle, and the SAR images for testing are acquired at a 30° depression angle. Similarly, the sample size of targets per class is not equal.

B. Classification Results and Analysis

Fig. 4 shows the classification performance of the proposed MKSFF-CNN during the SOC and EOC-D experiments in the form of the confusion matrix. The confusion matrix is the visualization tool used to evaluate the target classification performance. Its rows correspond to the true category labels of the target, while columns represent the predicted category labels of the target, and each element is the number of the corresponding target. It can be seen from Fig. 4(a) that the diagonal elements

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TABLE II
NUMBER OF TRAINING AND TEST IMAGES FOR THE EOC-D EXPERIMENTAL SETUP

Types	Training set		Test set	
	Depression	Number	Depression	Number
2S1	17°	299	30°	288
BRDM-2	17°	298	30°	287
T-72	17°	299	30°	288
ZSU-234	17°	299	30°	288

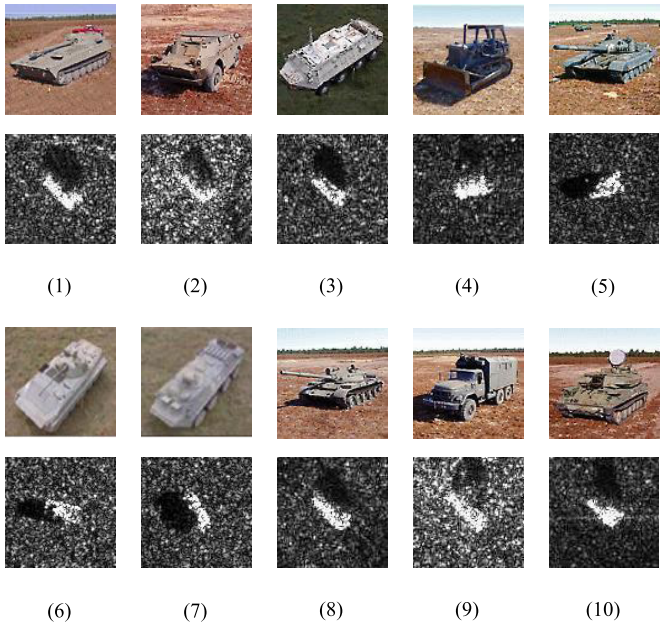


Fig. 3. Optical images (top) and corresponding SAR images (bottom) of various military targets in the MSTAR dataset: (1) 2S1, (2) BRDM-2, (3) BTR-60, (4) D7, (5) T-72, (6) BMP-2, (7) BTR-70, (8) T-62, (9) ZIL-131, and (10) ZSU-234.

in the confusion matrix of the SOC experiment are much larger than those in other positions. This means that the proposed MKSFF-CNN achieves high classification accuracy for each type of target in the SOC experiment, and the classification accuracy of T-72 even reaches 100%. In Fig. 4(b), the diagonal elements in the confusion matrix of the EOC-D experiment are also larger than those in other positions, but the values of some nondiagonal elements reach double digits, which means that a certain number of targets have been misclassified as other kinds. The classification performance of the proposed MKSFF-CNN in the EOC-D experiment is worse than that in the SOC experiment. This can be explained that the depression angles of the training set and the test set in the EOC-D experiment are quite different, resulting in lower classification accuracy.

In order to comprehensively validate the superiority of the proposed MKSFF-CNN, the article quotes a series of common SAR target classification methods to compare with the proposed MKSFF-CNN. These methods include CNN-SVM [32], A-ConvNet [12], CHU-Net [38], M-VDCNN [45], MSRIHL-CNN [40], VGG-Net [34], and Res-Net [35]. For each classification method, the relevant parameter settings are as follows.

1) For A-ConvNet [12], the convolutional layer with the kernels of 3×3 sizes is used to replace the fully connected layer to achieve data compression.
2) According to the settings in the original paper of [32], the kernel function of the SVM classifier used in the CNN-SVM classifier is the RBF function.
3) As for CHU-Net [38], the ensemble model is used, the layers in each unit are set to three, and the feature maps per layer are set to 34.
4) As for M-VDCNN [45], according to the settings in the original paper, since the DCNN with four views (4-VDCNN) achieves the best classification performance, it is selected for the classification performance comparison.
5) As for MSRIHL-CNN [40], the same parameters are used as in the original paper. The dimension of the reduced haar-like feature is 400, and the optimal fusion factor of γ is set to 0.3.
6) For VGG-Net [34] and Res-Net [35], VGG-16 and ResNet-50 are selected to test the classification performance.
7) For MKSFF-CNN, the number of kernels in the current convolutional layer is set to four times that in the previous convolutional layer. In this case, for the input SAR image, its length and width are reduced to half of the original after pooling, and then, its depth is expanded to four times the original after convolution, so the length of the extracted feature vector does not change.
8) For each method, the size of the input SAR image is 88×88. The stride size of all convolutional layers is set to one, and the stride size of all pooling layers is set to two. In addition, the learning rate is set to 10^{-4}, and the batch size and the epoch are, respectively, set to 20 and 100.

The metric of classification accuracy is used as the indicator to objectively evaluate the classification performance. The achieved values of the classification accuracy of various classification methods in the SOC experiment and the EOC-D experiment are shown in Tables III and IV, respectively. In Tables III and IV, it can be found that no matter in the SOC experiment or the EOC-D experiment, the classification accuracy of CNN-SVM is lower than that of the other methods. CNN-SVM uses the 13×13 kernel, which is so large for the SAR target in the MSTAR dataset, so that the local texture features of the SAR target are almost completely lost.

A-ConvNet uses a relatively deep feature extraction network to improve classification performance. However, it mainly uses a fixed kernel with a size of 5×5 for convolution, resulting in low feature representation completeness of the SAR target. A-ConvNet uses the convolutional layer instead of the fully connected layer to reduce the dimensionality of the data. Although the computing efficiency is improved, the classification accuracy is also affected to a certain extent.

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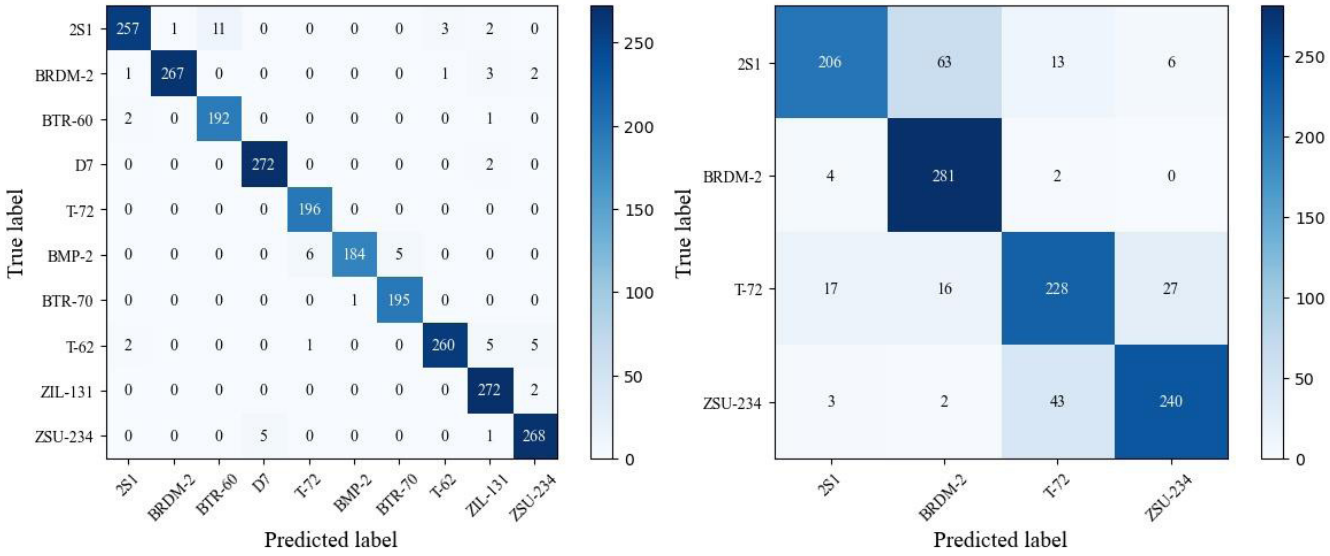


Fig. 4. Confusion matrix of the classification experiments obtained by MKSFF-CNN. (a) SOC experiment. (b) EOC-D experiment.

TABLE III
CLASSIFICATION ACCURACY OF VARIOUS TARGETS IN DIFFERENT METHODS IN THE SOC EXPERIMENT

Method	2S1	BRDM-2	BTR-60	D7	T-72	BMP-2	BTR-70	T-62	ZIL-131	ZSU-234	Total
CNN-SVM [32]	82.12	89.78	89.74	95.99	100	78.46	96.94	90.11	96.35	97.08	91.71
A-ConvNet [12]	83.94	91.61	91.79	97.08	98.98	83.08	98.47	94.87	97.08	95.26	93.24
CHU-Net [38]	78.10	92.34	94.87	96.72	98.98	82.56	94.39	94.87	99.27	94.16	92.62
4-VDCNN [45]	86.86	90.51	89.74	99.27	100	76.92	98.98	95.24	96.35	96.35	93.24
MSRIHL-CNN [40]	85.77	96.35	96.41	98.18	100	82.56	98.98	91.21	98.54	93.80	94.14
VGG-Net [34]	82.85	96.35	92.82	98.91	99.49	92.82	96.43	94.87	97.45	97.45	94.89
Res-Net [35]	81.75	90.51	93.85	98.54	99.49	94.36	97.96	91.21	96.72	96.72	93.81
MKSFF-CNN	**93.80**	**97.45**	**98.46**	**99.27**	**100**	**94.36**	**99.49**	**95.24**	**99.27**	**97.81**	**97.44**

CHU-Net uses a convolutional highway unit to effectively exploit information and compress overfitting, and it increases the width of the network by combining a convolutional highway unit with 3×3 kernels and a convolutional highway unit with 5×5 kernels so that the feature representation completeness is improved. However, the number of kernels used in each convolutional layer is relatively small, and the completeness of the feature representation will also be negatively affected.

Compared with CNN-SVM, the classification accuracy of 4-VDCNN is significantly improved. 4-VDCNN expands the view of input SAR image, and it extracts the feature information from multiple views for the same target. In addition, 4-VDCNN adopts a parallel CNN topology with multiple inputs. It gradually fuses the features learned from different views in different layers to increase the generalization ability of the network. However, 4-VDCNN takes a lot of time to expand the observation views, and the extracted multiview features depend on the imaging quality of the raw SAR images.

MSRIHL-CNN optimally fuses the deep features extracted by CNN and the local edge features extracted by the haar-like templates, and the fused features are fed into the SVM classifier for final classification. The deep features and the local edge features complement each other, providing more feature information for the final classification. The dimension of the local edge features extracted by the haar-like templates is very huge. It results in a large cost of the training time and limits the practical application of MSRIHL-CNN.

VGG-Net and Res-Net, as two popular CNN models, have good performance in both the SOC experiment and the EOC-D experiment. However, there are many parameters that need to be updated in VGG-Net and Res-Net, so a lot of time will be spent on network training. In addition, due to the limited sample size of the SOC and the EOC-D, and the high

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TABLE IV
CLASSIFICATION ACCURACY OF VARIOUS TARGETS IN DIFFERENT METHODS IN THE EOC-D EXPERIMENT

Method	2S1	BRDM-2	T-72	ZSU-234	Total
CNN-SVM [32]	38.19	94.77	70.83	82.99	71.68
A-ConvNet [12]	49.31	97.21	65.97	78.13	72.63
CHU-Net [38]	62.85	84.67	75.69	64.58	71.94
4-VDCNN [45]	44.44	93.38	70.83	82.29	72.72
MSRIHL-CNN [40]	63.89	96.86	73.96	63.89	74.63
VGG-Net [34]	61.11	89.20	77.78	83.33	77.85
Res-Net [35]	66.32	83.97	70.14	82.29	75.67
MKSFF-CNN	**71.53**	**97.91**	**79.17**	**83.33**	**82.97**

network complexity of VGG-Net and Res-Net, the networks of VGG-Net and Res-Net converge slowly during the training process.

The proposed MKSFF-CNN adopts the convolutional layer with the multichannel parallel topology. It uses kernels of different sizes to extract the multikernel-size deep features of the SAR target and fuses these features in an optimal way to minimize the loss. Moreover, MKSFF-CNN concatenates the fused features extracted by the convolutional layers of different dimensions and inputs them into the decision-making part of the network. MKSFF-CNN incorporates much more feature information for the final classification, and it elevates the feature representation completeness of the SAR targets, so it significantly improves the classification accuracy. More than that, its network structure is relatively simple, and it can achieve faster convergence. MKSFF-CNN achieves a large classification accuracy and a high efficiency, which has a high practical value.

Precision, recall, and $F1$-score [51] are common evaluation indicators for classification tasks. In classification tasks, for a certain class of samples, precision refers to the proportion of samples that really belong to this class in the samples that are judged to be this class, and recall is the proportion of correctly classified samples to the total number of such samples. $F1$-Score is the harmonic average of precision and recall. Models with higher $F1$-score are generally recognized with higher classification performance. The definitions of precision, recall, and $F1$-score are [47]

$$\text{Precision} = \frac{\text{TP}}{\text{TP} + \text{FP}} \tag{18}$$

$$\text{Recall} = \frac{\text{TP}}{\text{TP} + \text{FN}} \tag{19}$$

$$F1 - \text{Score} = 2 \cdot \frac{\text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}} \tag{20}$$

where TP is the number of the true positives, TN is the number of the true negatives, FP is the number of the false positives, and FN is the number of the false negatives.

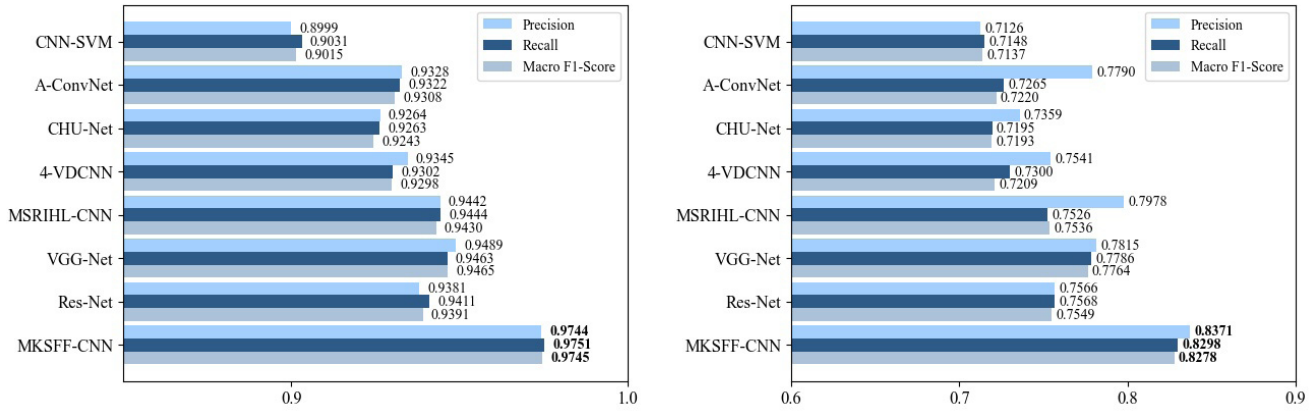
Macro $F1$-score [52] is the extension of $F1$-score in the multiclassification task. It is obtained by first calculating the precision, recall, and $F1$-score of each category in the multiclassification task, and Macro $F1$-score is achieved with the average value. The values of precision, recall, and Marco $F1$-score acquired by the classification methods are compared in Fig. 5. MKSFF-CNN achieves much higher values of precision, recall, and Marco $F1$-score whether, in the SOC experiment or the EOC-D experiment, it outperforms other methods quite a lot.

C. Classification Accuracy Evaluation Under Small-Size Training Datasets

In order to show that the proposed MKSFF-CNN can still maintain good performance even with small sample size, this part performs a series of comparative experiments based on an incomplete SOC dataset. A small part of the samples in each category of the SOC training dataset is randomly selected for training, and the classification results of different methods during each round are recorded. The acquired values of the classification accuracy under different numbers of training samples of each method are compared in Fig. 6.

It can be seen from Fig. 6 that no matter how the training sample size changes, MKSFF-CNN can always achieve higher accuracy than that of other methods. Even there are only 60 samples in each category of the training set, the classification accuracy of the proposed MKSFF-CNN can reach 88.29%, much higher than that of other methods. MKSFF-CNN performs better even under the case of small-size training samples; this can be explained by the simple network structure of the proposed MKSFF-CNN, which makes it easier to converge in the case of small samples, avoiding the problem of underfitting. Most importantly, MKSFF-CNN optimally fuses the feature information of different kernel sizes and layers for classification decisions, and it greatly improves the feature representation completeness. Undoubtedly, it can achieve a higher classification accuracy.

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(a) (b)

Fig. 5. Relevant indicators (including precision, recall, and Macro $F1$-score) of different classification methods. (a) SOC experiment. (b) EOC-D experiment.

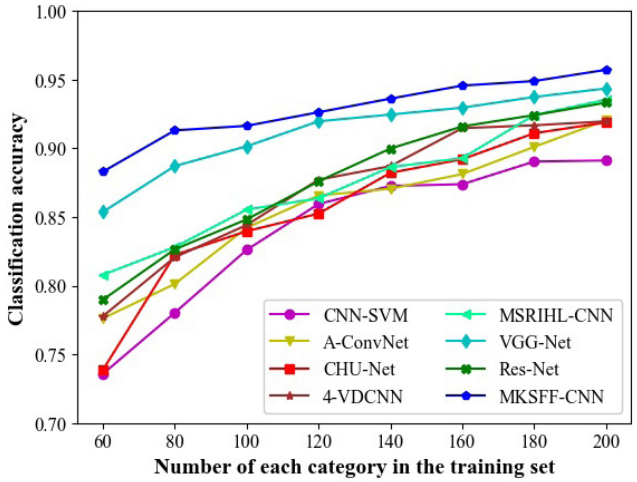


Fig. 6. Classification accuracy of different methods under small training sets in the SOC experiment.

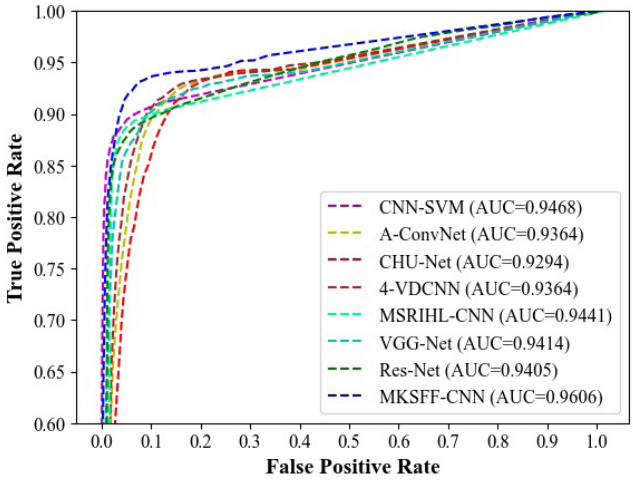


Fig. 7. ROC curves and the corresponding values of AUC of different classification methods in the SOC experiment.

D. Receiver Operation Characteristics (ROC) and Area Under the ROC Curve (AUC)

This part provides an objective evaluation of the classification performance in terms of ROC and AUC. ROC is an important metric for classification performance evaluation. It describes the relationship between the true positive rate (TPR) and the false positive ratio (FPR). The specific definitions of TPR and FPR are defined as follows:

$$\mathrm{TPR} = \frac{\mathrm{TP}}{\mathrm{TP} + \mathrm{FN}} \tag{21}$$

$$\mathrm{FPR} = \frac{\mathrm{FP}}{\mathrm{FP} + \mathrm{TN}}. \tag{22}$$

AUC is defined as the area under the ROC curve and the coordinate axis. It is usually used to assist the ROC to evaluate the performance of the classification method. Generally, the larger the AUC, the better the performance of the classification method will be.

A group of TPRs, FPRs, and corresponding AUCs are obtained by continuously altering the threshold of the classification decision during the testing process of the SOC experiment. With TPR as the ordinate and FPR as the abscissa, the ROC curve can be drawn by connecting these data. The ROC curves and the corresponding AUCs of the various classification methods based on the SOC experiment are shown in Fig. 7, from which it can be found that, compared with other classification methods, the proposed MKSFF-CNN can achieve a higher TPR while maintaining a lower FPR so that the corresponding value of AUC can reach the highest.

E. Computing Efficiency Evaluation

The computing time is an important indicator of the efficiency of each classification method. All the experiments are implemented by the Python codes running on the same computing station, which is composed of a six-core Intel Core i7-8750H CPU with 2.20-GHz frequency, a 16.0-GB memory, an Nvidia GeForce GTX 1060 GPU with 1.50-GHz frequency, 6.0-GB memory, and 1280 stream processors. The computing time is divided into training time and test time, as recorded in Table V.

The structure of CNN-SVM itself is relatively simple, but it uses an additional SVM classifier for classification, so it takes a little longer time, especially during the testing process. Compared to CNN-SVM, A-ConvNet has a deeper network, but it uses the convolutional layer instead of the

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TABLE V
COMPUTING TIME SPENT BY EACH CLASSIFICATION METHOD

Method	Training time (s)	Testing time (s)
CNN-SVM [32]	78.59	3.23
A-ConvNet [12]	57.92	1.20
CHU-Net [38]	97.84	1.74
4-VDCNN [45]	188.45	2.38
MSRIHL-CNN [40]	8421.19	2.43
VGG-Net [34]	429.54	4.01
Res-Net [35]	638.80	4.36
MKSFF-CNN	189.99	1.81

fully connected layer so that the parameters to be trained are reduced, and it does not need to train an additional classifier, so its computing time is less. Compared to the above two methods, the width of CHU-Net is larger, so its training time is longer. However, because it does not require an additional classifier for classification, its testing time is less than that of CNN-SVM. Compared to the above three methods, the structural complexity of 4-VDCNN is much higher since it incorporates the multiview information, so the computing time increases accordingly. MSRIHL-CNN uses an Adaboost classifier to reduce the huge dimensionality of the haar-like features; it consumes a lot of time during training, hundreds of times than that of A-ConvNet. The network depth of VGG-Net and Res-Net is higher than that of the other methods, so it takes them much longer time during training, several times than that of the other methods, and the testing time is also longer. The same as CNN-SVM, the proposed MKSFF-CNN adopts a relatively simple network model, and the time cost is much smaller than that of VGG, Res-Net, and MSRIHL-CNN. However, MKSFF-CNN additionally designs optimal fusion of the multikernel-size features and the multilayer features of the SAR target. It spends a longer training time than that of CNN-SVM, A-ConvNet, and CHU-Net. Since 4-VDCNN develops a parallel network for multiview feature learning, the time cost is equivalent to that of MKSFF-CNN.

F. Ablation Experiment

In order to objectively and comprehensively illustrate the superiority of the proposed MKSFF-CNN, the ablation experiment is designed. The ablation experiment is carried out based on the SOC experiment. The MKSFF and the multilayer feature concatenation are eliminated respectively from MKSFF-CNN to more clearly test their respective effect in SAR target classification performance. In addition, convolutional kernels of different sizes are used individually or in combination to extract feature information of SAR targets to test the effect of the size of the convolutional kernel used in the MKSFF on the SAR target classification performance. The results are recorded in Table VI.

TABLE VI
RESULTS OF THE ABLATION EXPERIMENT

Multi-kernel-size feature fusion		Multi-layer feature concatenation		Sizes of kernels used in convolution	Classification accuracy (%)
Yes	No	Yes	No		
	✓		✓	3×3	91.67
	✓	✓		3×3	92.25
	✓	✓		5×5	92.91
	✓	✓		7×7	92.49
	✓	✓		9×9	90.27
✓		✓		3×3 & 5×5	94.35
✓		✓		3×3 & 7×7	94.57
✓		✓		5×5 & 7×7	94.72
✓		✓		3×3, 5×5 & 7×7	**97.44**
✓		✓		3×3, 5×5, 7×7 & 9×9	95.05
✓			✓	3×3, 5×5 & 7×7	94.31

According to Table VI, whether the multilayer feature concatenation is eliminated or not, the MKSFF can help to improve the classification accuracy. Likewise, no matter whether the MKSFF is applied, the multilayer feature concatenation is conducive to the accurate classification of SAR targets. Moreover, when only one single fixed-size convolutional kernel (except the 9×9 kernels) is used for feature extraction, the classification accuracy obtained is about 92%. When two convolutional kernels of different sizes are combined and used for feature extraction, the classification accuracy obtained increases by about 2% to reach 94%. When convolutional kernels with sizes of 3×3, 5×5, and 7×7 are combined and used for feature extraction, such as the proposed MKSFF-CNN, the classification accuracy obtained increases by about 3% to reach 97%. However, whether the 9×9 kernel is used for feature extraction alone or participates in the combination of convolutional kernels of different sizes, it will cause a decrease in classification accuracy. It can be concluded from Table VI that both the MKSFF and the multilayer feature concatenation do elevate the completeness of the feature representation, thereby improving the performance of SAR target classification. In addition, the ablation experiment also proves that it is reasonable for the proposed MKSFF-CNN to choose the convolutional kernels with sizes of 3×3, 5×5, and 7×7 for feature extraction.

IV. CONCLUSION

A novel CNN model used for SAR target classification based on MKSFF-CNN is proposed in this article. MKSFF-CNN designs a convolutional layer with a

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multichannel parallel topology. It uses convolutional kernels of different sizes to extract the multikernel-size deep features of the SAR target, and these features are fused in an optimal way. Moreover, MKSFF-CNN concatenates the fused features extracted by the convolutional layers of different dimensions for the final classification. MKSFF-CNN elevates the feature representation completeness of the SAR targets, so the classification accuracy is improved significantly. Experiments carried out on the MSTAR dataset clearly illustrate that MKSFF-CNN can achieve better classification performance than other state-of-the-art methods, and its efficiency is high enough for practical engineering. It would be expected that MKSFF-CNN has broad application prospects in target classification tasks.

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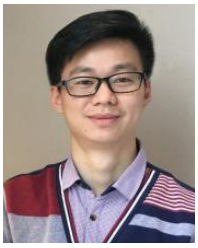
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